

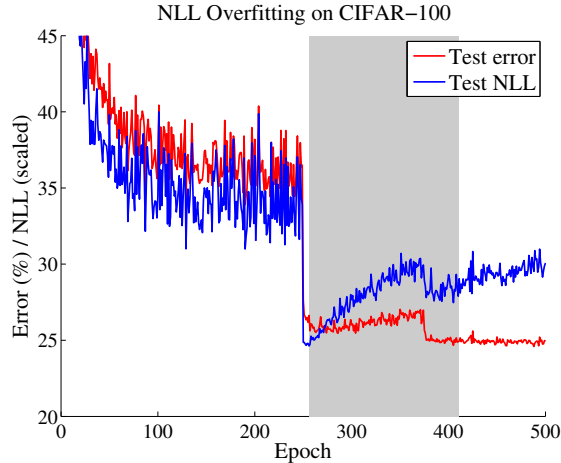


Figure 3. Test error and NLL of a 110-layer ResNet with stochastic depth on CIFAR-100 during training. NLL is scaled by a constant to fit in the figure. Learning rate drops by 10x at epochs 250 and 375. The shaded area marks between epochs at which the best validation loss and best validation error are produced.

den layers. Recent research suggests that these normalization techniques have enabled the development of very deep architectures, such as ResNets (He et al., 2016) and DenseNets (Huang et al., 2017). It has been shown that Batch Normalization improves training time, reduces the need for additional regularization, and can in some cases improve the accuracy of networks.

While it is difficult to pinpoint exactly how Batch Normalization affects the final predictions of a model, we do observe that models trained with Batch Normalization tend to be more miscalibrated. In the middle right plot of Figure 2, we see that a 6-layer ConvNet obtains worse calibration when Batch Normalization is applied, even though classification accuracy improves slightly. We find that this result holds regardless of the hyperparameters used on the Batch Normalization model (i.e. low or high learning rate, etc.).

Weight decay, which used to be the predominant regularization mechanism for neural networks, is decreasingly utilized when training modern neural networks. Learning theory suggests that regularization is necessary to prevent overfitting, especially as model capacity increases (Vapnik, 1998). However, due to the apparent regularization effects of Batch Normalization, recent research seems to suggest that models with less L2 regularization tend to generalize better (Ioffe & Szegedy, 2015). As a result, it is now common to train models with little weight decay, if any at all. The top performing ImageNet models of 2015 all use an order of magnitude less weight decay than models of previous years (He et al., 2016; Simonyan & Zisserman, 2015).

We find that training with less weight decay has a negative impact on calibration. The far right plot in Figure 2 dis-

plays training error and ECE for a 110-layer ResNet with varying amounts of weight decay. The only other forms of regularization are data augmentation and Batch Normalization. We observe that calibration and accuracy are not optimized by the same parameter setting. While the model exhibits both over-regularization and under-regularization with respect to classification error, it does not appear that calibration is negatively impacted by having too much weight decay. Model calibration continues to improve when more regularization is added, well after the point of achieving optimal accuracy. The slight uptick at the end of the graph may be an artifact of using a weight decay factor that impedes optimization.

NLL can be used to indirectly measure model calibration. In practice, we observe a *disconnect between NLL and accuracy*, which may explain the miscalibration in Figure 2. This disconnect occurs because neural networks can *overfit to NLL without overfitting to the 0/1 loss*. We observe this trend in the training curves of some miscalibrated models. Figure 3 shows test error and NLL (rescaled to match error) on CIFAR-100 as training progresses. Both error and NLL immediately drop at epoch 250, when the learning rate is dropped; however, NLL overfits during the remainder of training. Surprisingly, overfitting to NLL is beneficial to classification accuracy. On CIFAR-100, test error drops from 29% to 27% in the region where NLL overfits. This phenomenon renders a concrete explanation of miscalibration: the network learns better classification accuracy at the expense of well-modeled probabilities.

We can connect this finding to recent work examining the generalization of large neural networks. Zhang et al. (2017) observe that deep neural networks seemingly violate the common understanding of learning theory that large models with little regularization will not generalize well. The observed disconnect between NLL and 0/1 loss suggests that these high capacity models are not necessarily immune from overfitting, but rather, overfitting manifests in probabilistic error rather than classification error.

4. Calibration Methods

In this section, we first review existing calibration methods, and introduce new variants of our own. All methods are post-processing steps that produce (calibrated) probabilities. Each method requires a hold-out validation set, which in practice can be the same set used for hyperparameter tuning. We assume that the training, validation, and test sets are drawn from the same distribution.

4.1. Calibrating Binary Models

We first introduce calibration in the binary setting, i.e. $\mathcal{Y} = \{0, 1\}$. For simplicity, throughout this subsection,